



DS&AI LAB PROJECT | TERM MAY 2026

Course-Aware Personalized Learning Companion for IIT Madras BS Students

Milestone 1 - Problem Definition & Literature Review

Prototype course: Machine Learning Techniques (MLT), IITM BS Degree

Presented by

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Introduction & Context

Why a course-aware companion is needed



The Starting Point

- Students in the IIT Madras BS program learn primarily through recorded lectures, written notes, slide decks, previous year question papers (PYQs), FAQs, and discussion forums, each living on a different platform, with none aware of the others.
- Revising a single concept can mean searching a lecture video for the right timestamp, digging through a notes PDF, scrolling a forum thread for a similar doubt, and separately checking whether the concept has appeared in past question papers.
- General-purpose AI tools already help with parts of this, answering questions from uploaded documents, turning notes into quizzes, and are genuinely useful.
- What they do not do: stay aware of a specific course over time, remember what a specific student already struggles with, or connect a student's question back to the exact moment in a lecture where the answer was taught.
- This project is scoped strictly to the problem statement and proposed solution as defined for the project; claims about external tools are treated as general industry observations, not guarantees about current product behaviour, since AI products change quickly.

Project Objectives

From the foundational knowledge base to the full personalization loop



Primary Objective: Design and develop a prototype AI-powered learning companion for the IIT Madras BS Degree program that unifies scattered course resources into a searchable, source-grounded knowledge base and personalizes the learning experience based on each student's demonstrated understanding. The initial prototype is implemented using a single course: MLT.

Supporting Objectives

- ✓ Build a unified, course-specific knowledge base integrating lecture transcripts, notes, slides, PYQs, FAQs, and forum content.
- ✓ Provide source-grounded Q&A using Retrieval-Augmented Generation (RAG), so every answer traces back to specific course material.
- ✓ Enable lecture timestamp navigation, pointing to the exact point a recorded lecture explains a concept.
- ✓ Maintain a learner profile built from quiz performance and interaction history that persists across sessions.
- ✓ Detect knowledge gaps from the learner profile and turn them into concrete, actionable revision recommendations.
- ✓ Generate personalized quizzes targeted at a student's identified weak areas, rather than generic ones.
- ✓ Track learning progress over time so both student and system can see whether a gap is closing.
- ✓ Build an AI Question Intelligence Module to de-duplicate and cluster PYQs/FAQs/forum questions, presented via a Learning Analytics Dashboard.

Scope Boundary & Stakeholders

What the prototype covers, and who it serves, stakeholder list proposed



Scope Boundary

The problem statement is explicit that the initial prototype will focus on a single IIT Madras BS course: the team has selected

Machine Learning Techniques (MLT).

This is treated as a firm boundary: the objectives describe what the prototype should demonstrate for one course, not a claim of multi-course or program-wide coverage at this stage.

In scope: MLT lecture transcripts, notes, slides, PYQs, FAQs and forum threads.

Out of scope: other IITM BS courses; not evaluated as multi-course in this phase.

Expansion to additional courses is discussed later as an opportunity, not a committed objective for this phase.



Key Stakeholders



Students (primary)

IITM BS students taking MLT, direct end users of Q&A, quizzes, and the dashboard.



Instructor / TAs

Own the source material; gain visibility into common doubts and gap patterns across the cohort.



IITM BS Program Office

Indirect stakeholder, interested in learning outcomes and scalability to other courses.



Project Team

Owns build, evaluation, and iteration of the prototype through this course.

Literature Review

NotebookLM, adaptive quiz tools & general LLM assistants, compared against what this project needs

Dimension	NotebookLM	Adaptive Quiz & Assessment Tools	General-Purpose LLM Assistants
Grounding	<i>Source-grounded on uploaded docs, with citations</i>	<i>Grounded on uploaded notes for quiz generation only</i>	<i>Grounded mainly in pre-training data; hallucination risk</i>
Primary use case	<i>Summarizing & querying a fixed set of documents</i>	<i>Generating quizzes; adapting difficulty to performance</i>	<i>General conversational Q&A and explanation</i>
Course & learner awareness	<i>Limited to what's manually uploaded; no learner profile</i>	<i>Some platforms track performance within their own platform</i>	<i>None by default; not persistent unless a memory layer is added</i>
Lecture timestamp navigation	<i>Not designed for this</i>	<i>Not applicable</i>	<i>Not applicable</i>
Key gap vs. our objectives	<i>No PYQ/FAQ/forum integration or timestamp linking</i>	<i>No source-grounded, conversational Q&A</i>	<i>No course-specific grounding or persistent learner memory</i>

Qualitative comparison based on publicly documented capabilities - no formal benchmarking claimed.

Identified Gaps

Every category solves one piece well; none combine all of it



Course-specific grounding

No tool treats a course as a standing, evolving entity that lectures, notes, PYQs and forum activity all belong to.



Lecture navigation

None of the reviewed categories connect a specific question to the exact moment in a recorded lecture where it is addressed.



Assessment ↔ retrieval loop

Quiz performance doesn't automatically improve how the Q&A system prioritizes or explains that topic next time, and vice versa.



Question intelligence

PYQs, FAQs, and forum discussions accumulate duplicates over time; existing tools rely on manual curation to keep a question bank organized.



Persistent, course-tied learner profile

Where performance tracking exists, it's scoped to a tool's own environment/content, not tied to one course's actual structure and materials.

Opportunities

Read the other way, the gaps are what this project can pursue



Combine grounded retrieval with a living learner model

Pair RAG-based, source-grounded Q&A with a persistent learner profile, letting the two inform each other - a repeated question or a consistently-missed quiz topic can directly shape revision recommendations.



Use lecture timestamps as a genuine differentiator

Since no reviewed category is built around timestamp-level navigation, even a modest implementation offers something students can't currently get elsewhere - a text citation plus a lecture timestamp to verify and revisit a concept.



Turn scattered doubts into a structured, reusable asset

The AI Question Intelligence Module can convert PYQs, FAQs, and forum threads into a de-duplicated, concept-tagged repository - improving retrieval for every student, and surfacing common doubts to instructors.



Start narrow, prove the loop, then consider expansion

Because the prototype is scoped to MLT, the team can prove the full retrieval-plus-personalization loop end-to-end on a manageable dataset before considering extension to other IITM BS courses.



Learning Analytics as a shared view for students

A dashboard bringing together retrieval activity, quiz performance, and progress over time gives students one persistent view of their learning, currently split across whichever platform hosts each resource.

Evaluation Metrics & Benchmarks

How we plan to measure success on MLT



Proposed Evaluation Metrics

- **RAG Q&A:** Retrieval precision/recall-k $\geq 80\%$; answer faithfulness to source $\geq 88\%$; citation accuracy $\geq 85\%$
- **Lecture Navigation:** Timestamp accuracy within ± 10 seconds for $\geq 80\%$ of evaluated queries
- **Learner Profile:** Gap-detection precision $\geq 80\%$, validated against actual quiz performance
- **Personalized Quizzes:** Relevance to flagged weak areas $\geq 80\%$; pre/post revision score delta $\geq 15\%$
- **Question Intelligence:** Deduplication Precision $\geq 85\%$, Cluster F1 Score $\geq 80\%$
- **Overall Experience:** Task completion time $\geq 20\%$ faster than manual search; usability rating (survey) ≥ 3.8



Proposed Baselines

No existing tool combines all target capabilities, so no single formal benchmark applies. Three baselines are proposed instead:

1. Manual search baseline:

student manually searching materials: time & accuracy comparison.

2. Vanilla LLM baseline:

general-purpose LLM with no retrieval: isolates the value of grounding.

3. Upload-only RAG baseline:

NotebookLM-style tool, source-grounded but no learner profile.

Progress So Far

Milestone 1 complete, moving next into system design and data collection

Progress Snapshot



Problem statement finalized & scoped to MLT (Machine Learning Techniques)

Firm single-course boundary set, in line with the problem statement.



Literature review of NotebookLM, adaptive quiz tools & general LLM assistants

Each category assessed on grounding, course-awareness, learner modelling, and timestamp navigation.



Gap analysis, opportunity mapping & stakeholder identification completed

Five gaps and five opportunities documented; stakeholder groups drafted.



Evaluation metrics and baseline comparisons drafted for team review

Proposed, pending team sign-off before implementation begins.



Next: MLT dataset collection (lectures, notes, PYQs, FAQs, forum) & RAG pipeline design

First implementation step for Milestone 2.

Team Contributions: Milestone 1

All members contributed to review, discussion and finalization of this milestone report.

Member	Focus Area	Detailed Contribution
Mayank Singh	<i>Problem Definition, Scoping & Objectives</i>	Drafted the introduction framing the scattered-resource problem; authored the primary and 8 supporting objectives; defined and justified the single-course (MLT) scope boundary.
Ali Jawad	<i>NotebookLM Research & RAG Analysis</i>	Researched NotebookLM's source-grounded citation approach and feature set (summaries, audio overview, flashcards); identified its 3 structural gaps: course-awareness, learner modelling, timestamp linking.
Sachi Dhaturaha	<i>Adaptive Quiz Tools Review</i>	Surveyed adaptive assessment platforms and difficulty-adjustment approaches; compared their performance-tracking to our learner-profile objective; built the comparative summary (Table 1).
Aaryan Pratap Maurya	<i>General LLM Assistants Review</i>	Assessed grounding limitations and hallucination risk of general-purpose LLMs in education contexts; contrasted with RAG-anchored answering; co-developed the gap analysis.
Jibin V Mathews	<i>Opportunities, Conclusion & Compilation</i>	Synthesized opportunities from the 5 identified gaps; wrote the conclusion tying objectives to gaps; compiled, formatted and finalized the full milestone report and references.